Multimodal Classification using BERT and ResNet

Sayan Goswami

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1 Introduction

In many real-world applications, data exists in multiple forms such as text and images. Traditional machine learning models typically process only a single type of data. However, combining multiple data modalities can improve prediction performance.

This project implements a multimodal deep learning system that processes both textual and visual information to perform classification.

2 Objective

The objectives of this project are:

- To extract meaningful textual features using a pretrained BERT model
- To extract visual features using a pretrained ResNet50 model
- To fuse text and image features into a unified representation
- To train a classifier to predict the correct class
- To evaluate the performance of the multimodal model

3 Dataset

A small custom dataset was created consisting of image–text pairs belonging to two classes:

- Class 0: Dog images with corresponding text descriptions
- Class 1: Car images with corresponding text descriptions

Each sample in the dataset contains:

$$\text{Image} + \text{Text} + \text{Label}$$

4 Methodology

4.1 Text Feature Extraction

Text data is processed using the pretrained BERT model. The input sentence is tokenized and passed through BERT, and the embedding corresponding to the CLS token is extracted.

$$\text{Text} \rightarrow \text{BERT} \rightarrow \text{768-dimensional feature vector}$$

4.2 Image Feature Extraction

Images are processed using a pretrained ResNet50 model. The final classification layer is removed, and the model is used as a feature extractor.

$$\text{Image} \rightarrow \text{ResNet} \rightarrow \text{2048-dimensional feature vector}$$

4.3 Feature Fusion

The extracted text and image features are combined using concatenation:

$$768 + 2048 = 2816$$

This fused representation captures both semantic and visual information.

4.4 Classification

The fused feature vector is passed through a fully connected neural network layer:

$$\text{2816 features} \rightarrow \text{Linear Layer} \rightarrow \text{2 classes}$$

5 Training

The model is trained using the following configuration:

- Loss Function: Cross-Entropy Loss
- Optimizer: Adam
- Number of Epochs: 10
- Learning Rate: 0.001

It was observed that the training loss decreased consistently across epochs, indicating that the model successfully learned meaningful multimodal representations by combining textual and visual features.

6 Testing and Evaluation

After training, the model is evaluated on the dataset.

For each sample:

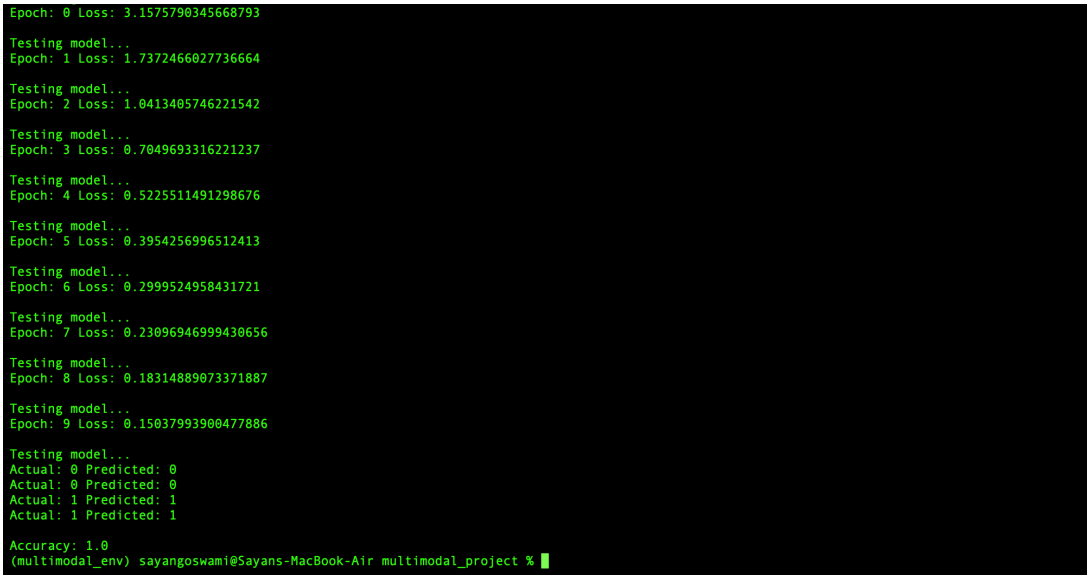
- Text and image features are extracted
- Features are fused
- The classifier predicts the class
- Prediction is compared with the actual label

Accuracy is computed as:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Samples}}$$

7 Results

The training process showed a consistent decrease in loss across epochs, indicating that the model successfully learned meaningful multimodal representations.



```
Epoch: 0 Loss: 3.1575790345668793

Testing model...
Epoch: 1 Loss: 1.7372466027736664

Testing model...
Epoch: 2 Loss: 1.0413405746221542

Testing model...
Epoch: 3 Loss: 0.7049693316221237

Testing model...
Epoch: 4 Loss: 0.5225511491298676

Testing model...
Epoch: 5 Loss: 0.3954256996512413

Testing model...
Epoch: 6 Loss: 0.2999524958431721

Testing model...
Epoch: 7 Loss: 0.23096946999430656

Testing model...
Epoch: 8 Loss: 0.18314889073371887

Testing model...
Epoch: 9 Loss: 0.15037993900477886

Testing model...
Actual: 0 Predicted: 0
Actual: 0 Predicted: 0
Actual: 1 Predicted: 1
Actual: 1 Predicted: 1

Accuracy: 1.0
(multimodal_env) sayangoswami@Sayans-MacBook-Air multimodal_project %
```

Figure 1: Training Loss and Model Predictions

From the results, it can be observed that:

- The loss decreases steadily from a high initial value to a low value.
- The model correctly predicts all samples.
- Final classification accuracy achieved is **100%**.

These results demonstrate the effectiveness of combining text and image features for classification.

8 Architecture Overview

$$
\begin{gathered}
\text{Text} \rightarrow \text{BERT (768)} \\
\text{Image} \rightarrow \text{ResNet (2048)} \\
\downarrow \\
\text{Feature Fusion } (768 + 2048 = 2816) \\
\downarrow \\
\text{Classifier (Linear)} \\
\downarrow \\
\text{Prediction (2 Classes)}
\end{gathered}
$$

9 Tools and Libraries

The implementation of the proposed multimodal model was carried out using Python. The following libraries were used:

- PyTorch for deep learning model implementation and training
- Pandas for dataset handling and preprocessing
- PIL (Python Imaging Library) for image loading
- Torchvision for image transformations and pretrained ResNet model
- HuggingFace Transformers for text feature extraction using BERT

10 Conclusion

This project successfully implements a multimodal deep learning system combining textual and visual features. The use of pretrained models allows effective feature extraction, and the fusion of these features enables accurate classification.

11 Limitations

- Small dataset size
- No separate test dataset (toy-model)
- Limited generalization capability